Purva Nalawade

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EDUCATION

Shah and Anchor Kutchhi Engineering College (University of Mumbai) — Mumbai, India

B.Tech in Cyber Security **(CGPA: 8.83/10)** · Honors in AI/ML **(CGPA: 9.5/10)** — *2023 – Expected May 2027*

Events Lead & Technical Team Member, Cyber Council · Events Head, EBSB Club · Operations Team Member, GDG on Campus

Relevant Coursework: AI/ML with Cyber Security, Cryptography & Network Security, Cyber Crime Investigation & Digital Forensics, Security Operations Center, Penetration Testing, Cloud & Multi-Cloud Security

PROFESSIONAL EXPERIENCE

Maharashtra Cyber | *Cybersecurity Intern* — *Jan 2026 – Aug 2026*

- Traced IP origins and mapped digital footprints for 50 NCMEC-referred cases daily over 4 months, using the case portal and IP geolocation tools to identify the originating district/city and forward findings to the relevant local law enforcement unit.
- Tracked extortion and deepfake dissemination across Instagram, WhatsApp, and Snapchat using OSINT tools (Maltego, ExifTool, DarkOwl, WHOIS, theHarvester), supporting takedown requests on 15 cases.
- Triaged 20-30 IBM QRadar (SIEM) alerts daily, escalating suspicious activity to senior analysts for investigation.

CyberPeace Foundation – Centre of Excellence | *Research Intern* — *Jan 2026 – May 2026*

- Conceived and built modular pipelines for AI-AwareX, integrating a Differentiable JPEG Simulator via Projected Gradient Descent (PGD) to ensure adversarial cloaking survives platform re-encoding.
- Implemented deterministic heuristic and regex pipelines for PII redaction (Aadhaar, PAN, financial identifiers) and AI-generated text classification.

DeepCytes Cyber Labs | *Red Team Analyst – Intern* — *Jul 2025 – Dec 2025*

- Executed web application penetration testing (VAPT) across live target environments using Burp Suite, Nmap, and custom automation scripts.
- Identified and exploited critical vulnerabilities aligned with OWASP Top 10 (including SQL Injection, IDOR, and broken authentication); delivered remediation PoCs and executive risk ratings.

Millennium Enterprise | *Full Stack Development Intern* — *Mar 2025 – May 2025*

- Co-developed the Millennium Enterprises book publishing website using Node.js, Express, and MongoDB, implementing JWT/OAuth authentication schemes and strict input validation layers to mitigate XSS and injection risks.

PROJECTS

- **GuardrailX – LLM Governance & Safety Framework** — *Python · PyTorch · HuggingFace · LLM Red Teaming*
  Automated adversarial testing suite evaluating structural safety degradation across Llama 3.3 70B, Mistral, and Qwen. Implemented automated refusal-delta scoring and real-time input/output policy enforcement shields.
- **CCTV Cyber Defense – Zero-Trust Surveillance Hardening** — *Python · RTSP · MediaMTX · Suricata (IDS/IPS) · VPN/TLS*
  Consolation Prize MHA, BPR&D & NCRB Hackathon 2.0 (New Delhi)
  Implemented a zero-trust architecture for video surveillance streaming via MediaMTX and encrypted RTSP feeds; deployed active IDS/IPS anomaly detection to mitigate camera node hijacking and MITM vectors.
- **Phishing Link Detection & Takedown Platform** — *Next.js · Django · Supabase · CERT-In Integration*
  Top 10 Finalist – National iDEA Hackathon 2025
  Automated detection and escalation pipeline for public sector banking domains, validating SSL anomalies and coordinating real-time takedowns with national incident response frameworks.

RESEARCH & PUBLICATIONS

- **NIDS Adversarial Transferability** – WorldS4 2026 (Springer LNNS, Scopus-indexed), *accepted, oral presentation Jul 28, 2026*. Cross-dataset transferability of adversarial attacks (BoundaryAttack via ART) against RF/XGBoost on CICIDS2017 & UNSW-NB15.
- **TrustFade: LLM Inference-Time Forgetting** – ICTCS 2026 (Springer LNNS), *accepted, oral presentation Dec 17, 2026*. 648-trial study of behavioral suppression across Llama model scales and probe types.
- **Deceiving AI While Preserving Human Perception: A Generative Adversarial Approach to Facial Identity Protection** – Accepted at AIS 2026 (Springer LNCS; withdrawn prior to publication); currently under review at ICISS 2026.
- **Indian Patent Application No. 202621074738** (filed Jun 16, 2026, Published, Under Examination) – Cybersecurity Defense System and Method for Zero-Knowledge Adversarial Forensic Simulation and Vulnerability Assessment.
- Editorial team member & contributor, *Intelligent Forensic* (Book published by Millennium Enterprises).

CERTIFICATIONS & COMPETITIVE ACHIEVEMENTS

- **Certifications:** Certified Ethical Hacker (CEH v13), CompTIA Security+ (SY0-701), Certified in Cybersecurity (CC – ISC2), AWS Academy Graduate (Cloud Architecting, Cloud Security Foundations, Cloud Operations, Cloud Developing).
- **TryHackMe:** Top 2% Globally (Handle: 404xShadow) | Junior Penetration Tester & Web Fundamentals.
- **Hackathons:** Consolation Prize Winner (Top 12 / 750+ Teams Nationwide) – Ministry of Home Affairs, Bureau of Police Research and Development & National Crime Records Bureau, Ministry of Home Affairs Hackathon 2.0 (New Delhi); Top 10 Finalist – National iDEA Hackathon 2025.
- **Competitive CTFs:** DawgCTF 2025 (UMBC, USA – 64th), KJSSE CTF 3.0, Bypass CTF 2025 (AIT Pune), CREST CTF 2026, Hacker's Gambit (OWASP).

TECHNICAL SKILLS

- **Core Languages:** Python, C, Bash, SQL, JavaScript, TypeScript
- **Security & Systems:** FlareVM, QRadar (SIEM), Burp Suite, Suricata, Wireshark, Nmap, Metasploit, Autopsy
- **AI / ML Security:** PyTorch, HuggingFace Transformers, ART (Adversarial Robustness Toolbox), scikit-learn, Presidio, spaCy
- **Cloud & Infrastructure:** AWS (Cloud Architecture, Security & Operations), Linux (Kali/Ubuntu), Docker, VirtualBox/VMware, Git & GitHub, GitHub Actions (CI/CD)